

Miles Jennings

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EDUCATION

University of California Irvine

Irvine, CA

B.S Computer Engineering — Deans Honor List

Sept. 2022 – June 2026

Relevant Coursework: Organization of Digital Computers (Computer Architecture) Electronics I, Electronics II, Network Analysis I, Network Analysis II, Principles of Digital Design, Processor Hardware/Software Interfaces, Organization of Digital Computers

Organizations: HyperXite, FUSION, FASAE

EXPERIENCE

Test Engineer Intern

Buena Park, California

Leach International Corp.

March. 2026 – Present

- Troubleshoot, made documentation for, and repaired test equipment by analyzing existing system architectures and adapting to legacy setups, eliminating production bottlenecks and ensuring on-time delivery
- **Developed** a **Python**-based capacitor test/conditioning system (LCR6002, High Voltage Power Supply, Safety Analyzer) for validation of **490** components, eliminating outsourced testing and reducing related company costs by **60%**
- Built and maintained hardware test setups using oscilloscopes, multimeters, and power supplies to perform functional and performance testing on electrical components

Control Systems Engineer

Irvine, California

HyperXite – SpaceX Hyperloop

Aug. 2025 – Present

- Designed, tested, and optimized STM32H753ZI-based control hardware in **C/C++**, interfacing with sensors via **UART, I2C, and CAN** while evaluating components for performance and cost improvements
- Built ESP32-based bidirectional communication system using ESP-NOW for real-time control and telemetry between pod and station and utilizing FreeRTOS to schedule critical tasks
- Ensured hardware–software compatibility by defining system constraints and key telemetry parameters based on feedback from team surveys identifying the most critical data points to track in the GUI

P2S Technology Intern

Long Beach, California

P2S Engineering

June 2025 – March 2026

- Assisted in the design and documentation of low-voltage and IT infrastructure systems for large scale buildings
- Improved design accuracy and project timelines by reducing regulatory revision cycles by **11%** through proactive 3D modeling and cross-functional coordination
- Collaborated with multidisciplinary teams to coordinate technology system requirements with electrical and mechanical designs to ensure reliable infrastructure design

PROJECTS

AudioVisor | C, I2C, PDM, nRF52840, TypeScript, Vite, Supabase

Sept. 2025 – March 2026

- Led a team of 5 in designing and prototyping an audio-reactive wearable system, guiding embedded hardware integration, component selection, and real-time signal processing architecture
- Built an audio-reactive visor using an nRF52840, utilizing **C** to receive and handle digital microphone input through **PDM** communication for real-time sound magnitude/direction detection
- Developed a real-time web dashboard for the AudioVisor system to visualize and analyze audio data, achieving **84%** accuracy in detecting sound direction and amplitude during initial trials

Building Management System | Python, Raspberry Pi, LCD, DHT-11

June 2025

- Designed and constructed a fully functional Building Management System (BMS) on Raspberry Pi using real-time data from DHT-11 and PIR sensors to control HVAC, ambient lighting, fire alarms, and security systems
- Programmed an interactive LCD interface and logging system to display live status updates, trigger event-based alerts, and ensure time-stamped logs with synchronization and interrupt-driven input handling

TECHNICAL SKILLS

Languages: Python, C/C++, Java, Verilog, RISC-V, MIPS

Software/Tools: LabVIEW, Revit, AutoCAD, KiCad, Bluebeam, Arduino IDE, VSCode, Vivado, LTSpice, Cadence, Linux, Git, GitHub, STM32CubeIDE, MobaXterm

Hardware: Oscilloscope, Power Supply, Multimeter, Circuit Design & Analysis, Microcontrollers, ESP32, Embedded Systems, PCB Design, Arduino, Power Supplies, FPGA